

Analysis of Bottleneck in a Manufacturing Plant

Group Number: 9

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Name of Simulation Tool: FlexSim

Brief Description of the features of FlexSim

FlexSim is a powerful 3D simulation and modeling software designed to help engineers, researchers, and decision-makers visualize, analyze, and optimize complex systems and processes. It allows users to create virtual replicas of real-world environments such as manufacturing systems, supply chains, healthcare facilities, logistics operations, and service systems to test and improve their performance in a risk-free digital environment.

Key Features of FlexSim

1. **3D Simulation & Visualization** – Build realistic 3D models to clearly visualize workflows, detect bottlenecks, and communicate ideas effectively.
2. **Discrete Event Simulation (DES)** – Models systems as a series of time-based events for accurate process analysis and optimization.
3. **Ease of Use & Flexibility** – Drag-and-drop modeling with optional FlexScript coding for customization and advanced logic.
4. **Data Integration** – Connects with Excel, databases, and real-time systems for data-driven simulation and digital twin applications.
5. **Statistical Analysis & Reporting** – Provides detailed charts, dashboards, and metrics like throughput, utilization, and wait times.
6. **Optimization Tools** – Includes Experimenter and Optimizer modules for scenario testing and performance improvement.
7. **Industry Applications** – Widely used in manufacturing, logistics, healthcare, and service systems for layout and process optimization.
8. **Collaboration & Communication** – Offers visual outputs and easy model sharing for better teamwork and stakeholder understanding.

Problem Definition

A multi-stage processing system is consistently failing to meet its desired daily output quota. This inefficiency results in higher operational costs, missed deadlines, and an inability to meet customer demand.

While the overall system output is visibly low, management is unsure which specific part of the process is causing the slowdown. The problem is to **identify the specific station or resource (e.g., machine, operator, or buffer) that is the primary bottleneck** and quantify its impact, as this constraint limits the entire system's throughput.

Problem Description and Schema diagram used in Simulation

The system to be modeled in FlexSim is a discrete-part manufacturing line, simulated over a **48-hour (2,880-minute)** operational period. The objective is to analyze the system's throughput, identify any bottlenecks, and examine resource utilization.

Sr. No.	Task Executors/Resources	Parameters
1	Source	Input is unlimited
2	Lathe	QTY: 1 at a time and process time: 10 mins
3	Queue	QTY: 1 capacity 1000
4	Drill	QTY: 1 at a time and process time: 5 mins
5	Queue	QTY: 1 capacity 1000
6	CNC m/c	QTY: 1 at a time and process time: 7 mins
7	Queue	QTY: 1 capacity 1000
8	Mill	QTY: 1 at a time and process time: 15 mins
9	Transporter	QTY: 1 at a time
10	Sink	Output

The problem involves analyzing a multi-stage manufacturing process to identify and resolve production bottlenecks. The "before" layout, which represents the current state, models a system where parts are generated by a **Source**. These parts first go to a **Lathe**, which requires **Operator1** to function. After the lathe, parts may wait in a small buffer, **Queue1**, before being processed by the **Drill**, which is attended by **Operator2**.

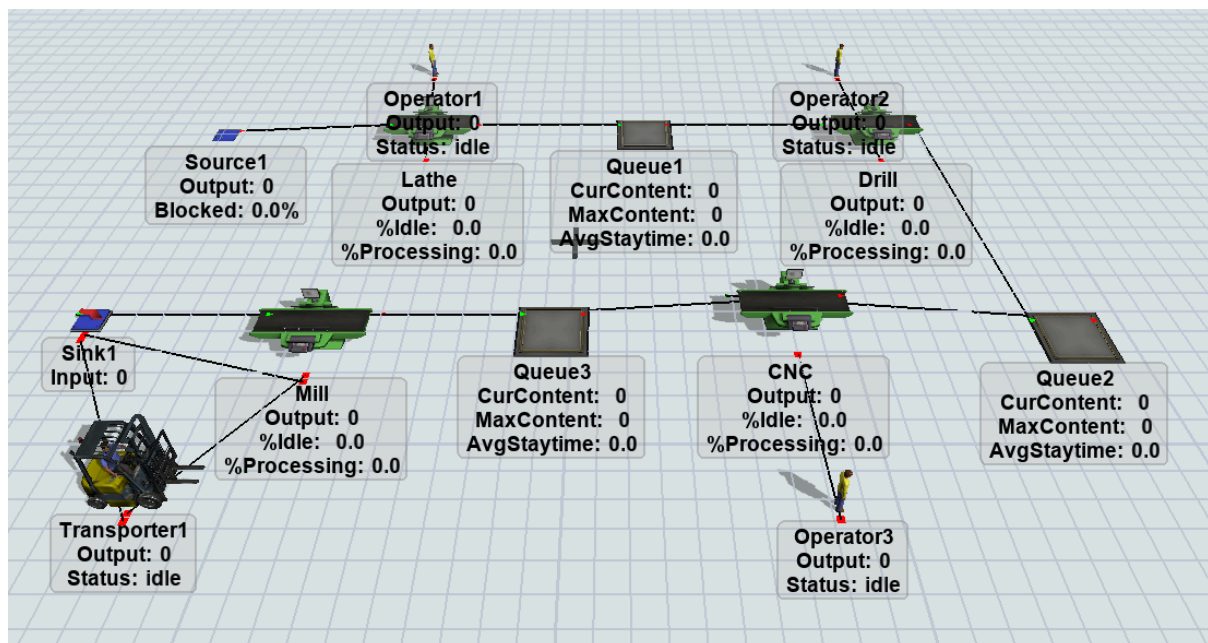
Following the drilling operation, parts enter another buffer, **Queue2**, to wait for the **CNC** machine. Once processed by the CNC, all parts are routed to a single, large-capacity buffer, **Queue3**, to wait for the final processing step at the **Mill**. After the Mill completes its work, the finished parts are sent to the **Sink**, which represents the end of the line. The entire system is simulated over a 2-day (2880-minute) period to gather performance statistics. The initial schema suggests a potential imbalance, as the line converges from multiple machines into a single Mill at the very end, which is the primary focus of the investigation.

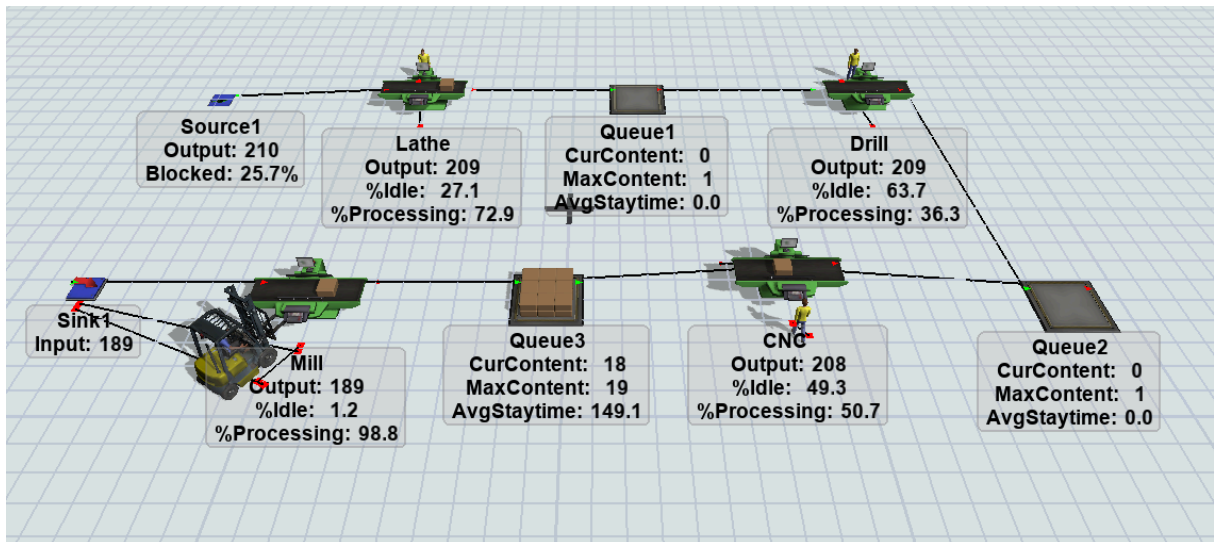
Simulation Objective

The primary objective of this simulation study is to perform a bottleneck analysis on the existing manufacturing layout. The goal is to use the 2-day simulation data to identify the specific resource (machine or operator) that is constraining the overall throughput of the system. This is achieved by examining key metrics like machine utilization, queue lengths, and part waiting times. A secondary objective is to propose and test a viable solution to alleviate the identified bottleneck. Finally, the simulation aims to quantify the improvement by comparing the "before" model's output against the "after" model's output, thereby validating the effectiveness of the proposed change.

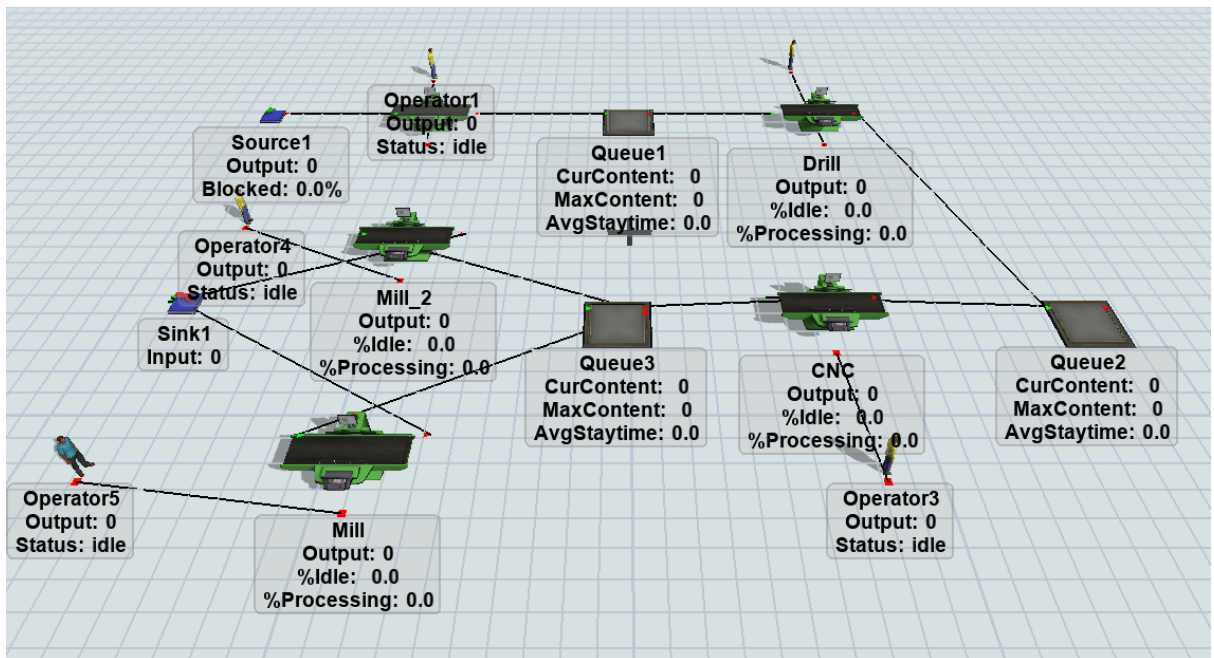
Screenshots of Simulation

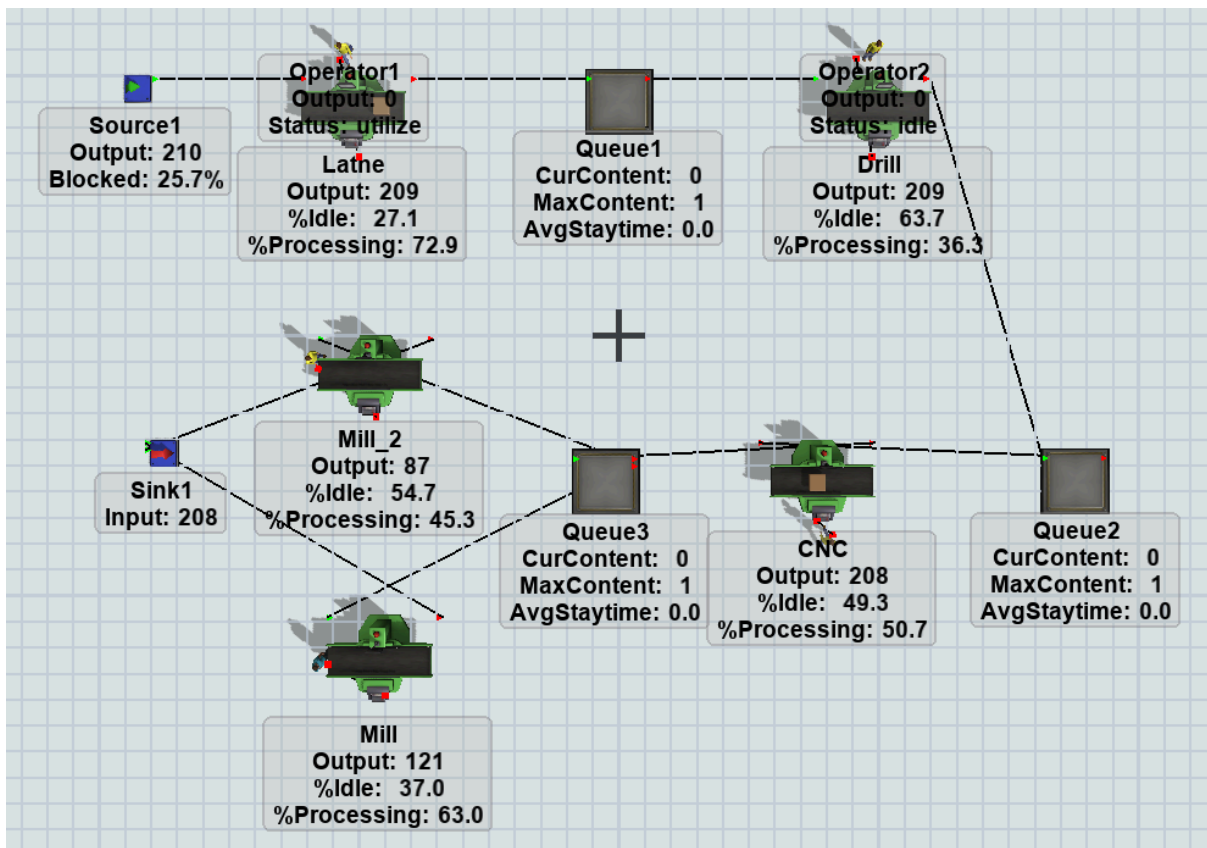
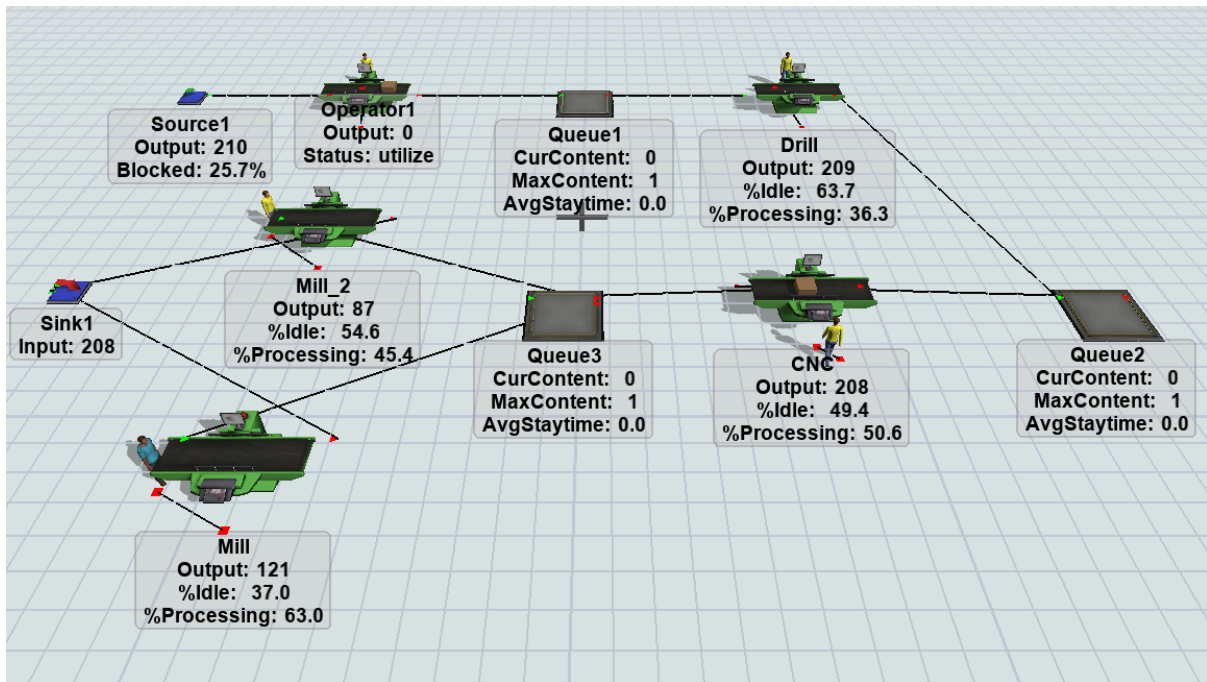
Before optimization:





After optimization:





Inferences from the Simulation

- The "before" model clearly reveals a critical bottleneck at the Mill station, which is overwhelmed at **98.8%** processing time.
- This bottleneck creates a severe backup in Queue3, which has an average wait time of **149.1 minutes** and is filled to its maximum capacity.

- The "after" model's solution, which involved adding a parallel Mill_2, successfully resolved this bottleneck by splitting the workload from the CNC.
- This solution increased total system throughput by **10.1%** (from 189 to 208 units) and completely eliminated the wait time and backup at Queue3.
- With the Mill bottleneck fixed, the simulation now shows that the new system constraint is the Lathe, which (at **72.8%** utilization) is responsible for the **25.7%** block rate at the Source.

Results:

Parameters	Before	After
Total Parts at Source	210	210
Total Parts at Sink	189	208
Average time required to dispatch the part	2880/189	2880/208
Average work in progress	210-189	210-208
Bottleneck occurring at	Mill	Lathe

2 Alternative Simulation Tools for the same Problem Statement

Anylogic

AnyLogic is a versatile, multi-method simulation software that supports Discrete Event Simulation (DES), Agent-Based Modeling (ABM), and System Dynamics (SD). It is widely used for modeling complex manufacturing, logistics, and supply chain systems. Its ability to integrate different simulation paradigms makes it ideal for analyzing both machine-level operations and human or logistical factors simultaneously. AnyLogic also offers excellent visualization features and built-in optimization tools, allowing users to test multiple scenarios and select the most efficient configuration for process improvement.

MATLAB/Simulink

MATLAB/Simulink provides a powerful platform for modeling, simulating, and analyzing dynamic systems through its block-diagram environment. It is especially effective for systems involving control mechanisms, automation, and signal processing, making it suitable for simulating manufacturing operations with embedded control logic. With its SimEvents toolbox, MATLAB enables discrete event simulation similar to FlexSim, allowing users to track process flow, resource utilization, and time-based events. Additionally, its strong

integration with data analytics and optimization toolboxes allows for advanced performance evaluation and parameter tuning.

Conclusion

This simulation project was highly effective in diagnosing and solving a major production-line constraint. The "Before" model definitively proved that the Mill was the single point of failure, limiting plant output and causing excessive work-in-process inventory to build up. The "After" model successfully validated the proposed solution of investing in a second, parallel Mill_2. This change not only fixed the primary issue but also resulted in a significant 10.1% increase in overall throughput. The project highlights the value of simulation for providing data-driven evidence to justify capital expenditures. It also correctly demonstrates the iterative nature of continuous improvement, as solving the Mill bottleneck has now revealed the Lathe as the next, less severe constraint to target for future optimization.